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Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Time: 03 Hours

Code: 103402 ***Subject: Electrical Machine - II*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[2x7=14]**

(a) Increasing the air gap in an induction motor generally leads to:
- (i) Increased magnetic flux and improved efficiency
- (ii) Reduced magnetizing inductance and a lower power factor
- (iii) An increase in rotor speed
- (iv) A higher starting torque

(b) In V/f control (voltage-to-frequency control), if both voltage and frequency are reduced while keeping the ratio constant, the maximum torque of the motor:
- (i) Increases significantly
- (ii) Decreases significantly
- (iii) Remains approximately the same
- (iv) Depends solely on rotor resistance

(c) Hysteresis motors are particularly suited for applications requiring:
- (i) High starting torque under heavy loads
- (ii) Precise speed control with low ripple
- (iii) Rapid acceleration and deceleration
- (iv) High torque density for industrial drives

(d) One disadvantage often encountered with reluctance motors is:
- (i) High maintenance due to brush wear
- (ii) Complex and noisy operation
- (iii) Requirement for external excitation
- (iv) Excessive starting current

(e) The term "step angle" in a stepper motor refers to:
- (i) The angle between the stator poles
- (ii) The maximum rotation of the motor at full load
- (iii) The angular displacement per input pulse
- (iv) The phase shift between different windings

(f) In a brushless DC motor, electronic commutation is performed by:
- (i) A built-in commutator and brushes
- (ii) An external electronic controller
- (iii) The natural inductance of the stator
- (iv) Permanent magnets on the stator

(g) Short-pitch (or chorded) windings are used in AC machines primarily to:
- (i) Increase the induced fundamental EMF
- (ii) Reduce the harmonic content (with a slight reduction in fundamental EMF)
- (iii) Simplify the winding process
- (iv) Increase the number of coils per phase

(h) Compared to a distributed winding, a concentrated winding:
- (i) Has a higher winding factor
- (ii) Produces an EMF with a more sinusoidal waveform
- (iii) Exhibits a lower distribution factor, leading to higher harmonic content
- (iv) Is always used for high-power applications

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(i) At the minimum point on the V-curve of a synchronous machine, the armature current is minimum because:
   (i) The machine is under-excited
   (ii) The machine is over-excited
   (iii) The armature reaction is minimized (approximately balanced), leading to nearly unity power factor
   (iv) The field current is zero

(j) When two alternators are paralleled. if one is over-excited and the other under-excited. then:
   (i) Both will share active load equally regardless of excitation
   (ii) The over-excited unit will supply reactive power while the under-excited unit will absorb reactive power
   (iii) The alternators will not be able to share load at all
   (iv) Only the over-excited alternator will supply both active and reactive power

Q.2 Solve both questions : **[14]**

(a) Draw the torque-slip characteristic of three phase induction motor and show the suitable derivations that maximum torque is independent of rotor resistance.

(b) A 6-pole 50Hz induction motor has a rotor resistance of 0.03 Ω per phase and standstill reactance of 0.4 Ω per phase. Determine the speed at which the maximum torque is developed.

Q.3 Solve both questions : **[14]**

(a) Explain the working of single-phase induction motor using double revolving field theory.

(b) Explain construction and working of any type of capacitor split phase induction motor. What type of capacitor are used.

Q.4 Solve both questions : **[14]**

(a) What are the similarities between brushless DC motor and conventional DC motor. Give the advantage of brushless DC motor over conventional DC motor.

(b) Explain the construction, working and application of stepper motor.

Q.5 Solve both questions : **[14]**

(a) The field excitation of three phase synchronous motor operating on fixed load is increased from a very low value with the help of a V curve.

(b) Explain the two-reaction theory applicable to salient pole synchronous machine.

Q.6 Solve both questions : **[14]**

(a) Explain how the desirable feature of high starting torque and low operating slip are obtained in a double cage polyphase induction motor.

(b) Why is starter necessary for starting of induction motors? Name different starting methods for 3-phase induction motor.

Q.7 Solve both questions : **[14]**

(a) Draw the physical arrangement of winding in stator and cylindrical rotor.

(b) Describe pulsating magnetic field of ac in winding with spatial displacement.

Q.8 Solve both questions : **[14]**

(a) Draw and explain the phasor diagram of loaded synchronous generator under lagging, leading and unity power factor condition.

(b) Calculate the rms value of induced EMF per phase of a 10 pole three phase 50 Hz alternator with 2 slots per pole per phase and 4 conductors per slots in two layers. The coil span is 150 degrees. The flux per pole has a fundamental component of 0.12 Wb and a 20% third harmonic component.

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Q.9 Write short notes on any two of the following: - **[7x2=14]**

(a) Cogging and crawling

(b) Parallel operation of alternators

(c) Pitch factor and Distribution factor

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